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Sergi Gomez; Rodolfo Jun Belen; Mark Kiehlbauch; Eray S. Aydil



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Etching of high aspect ratio structures in Si using SF₆/O₂ plasma

Sergi Gomez and Rodolfo Jun Belen

Department of Chemical Engineering, University of California, Santa Barbara, California 93106

Mark Kiehlbauch

Lam Research Corporation, 4400 Cushing Parkway, Fremont, California 94538

Eray S. Aydil^{a)}

Department of Chemical Engineering, University of California, Santa Barbara, California 93106

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We have investigated etching of deep ($\sim 10\ \mu\text{m}$) submicron diameter holes with high aspect ratios (>10) using plasmas maintained in mixtures of SF₆ and O₂ gases. The etching experiments were conducted in a low-pressure (5–80 mTorr), high-density, inductively coupled plasma etching reactor with a planar coil. We have studied the effects of pressure, rf-bias voltage, and SF₆-to-O₂ gas ratio on the etch rate, selectivity, and feature profile using Si wafers patterned with 0.35–0.5 μm diameter holes in a SiO₂ mask. Visualization of the profiles with scanning electron microscopy is used in conjunction with plasma diagnostics such as optical emission and mass spectroscopies to understand the key factors that control the anisotropy, selectivity, and etch rate. The F-to-ion flux ratio and F-to-O flux ratio are found to be the important plasma parameters that determine the etch rate and anisotropy. Increasing the SF₆-to-O₂ ratio in the feed gas increases the F-to-O ratio in the plasma. At high SF₆-to-O₂ ratio, the mask undercut is severe because sidewall passivation by O atoms cannot keep up with the chemical etching by F atoms. As the F-to-O ratio is decreased, effective sidewall passivation by O atoms results in nearly vertical sidewalls. A further reduction in the F-to-O ratio results in sidewalls that slope inwards toward the bottom of the feature. © 2004 American Vacuum Society. [DOI: 10.1116/1.1710493]

I. INTRODUCTION

Plasma etching of high aspect ratio (depth-to-width) structures in Si is an important step in manufacturing capacitors for memory devices, and integrated components for microelectromechanical systems (MEMS).^{1–3} During manufacturing of integrated circuits (ICs) and MEMS devices, features such as holes and trenches with aspect ratios ranging from 10 to several 100 are etched in Si. For example, trenches for state-of-the-art capacitors for memory devices are less than 0.1 μm wide and 1–10 μm deep. In contrast, in MEMS devices, trenches and holes are 1 to 100 microns wide and hundreds of microns deep.⁴ Despite the difference in feature sizes between ICs and MEMS devices, the scientific and engineering challenges for etching high aspect ratio structures (HARS) are similar. A successful HARS etching process requires high etch rates without sacrificing feature anisotropy and selectivity to the mask material. In addition, etching must be uniform in both macroscopic (over a wafer, $\sim 300\ \text{mm}$) and microscopic (across several features, $\sim 1\text{--}10\ \mu\text{m}$) length scales with excellent wafer-to-wafer reproducibility.

Fluorine-containing plasmas generated from gases such as CF₄, C₄F₈, NF₃, and SF₆ are commonly used in HARS applications because F etches Si chemically with very high etch rates.^{1,5–10} However, rapid chemical etching of Si by F also results in significant lateral etching, which compromises the anisotropy.^{6,9,11} To inhibit lateral etching, the feature

sidewalls must be protected by a passivating layer. For example, O₂ is usually added to SF₆ plasmas to form a thin fluorinated silicon oxide passivating layer on the sidewalls, resulting in anisotropic etching.^{3,5,7,8,11,12} Etching proceeds only at the bottom of the feature, where ions bombard the surface continuously, removing any passivation layer and exposing fresh Si to F atoms. In SF₆/O₂ plasma etching, substrate temperature may also be lowered to further inhibit lateral etching.^{3,13–15} Plasma etching processes based on mixtures of SF₆ and O₂ have emerged as an alternative to the time-multiplexed Bosch process,¹ which uses alternating passivation and etching steps to produce high aspect ratio features. Several investigators have studied etching of Si using SF₆/O₂ plasma.^{3,4,7,8,13–16} For example, Boufnichel *et al.* and Aachboun *et al.* investigated the effect of process parameters on Si etching at cryogenic temperatures and reported time-averaged etch rates of $\sim 5\ \mu\text{m}/\text{min}$ for etching 1.2–100 μm -wide deep trenches with aspect ratios in excess of 10 in an inductively coupled plasma reactor using 5%–10% O₂ in SF₆.^{4,13–15} Local bowing immediately under the mask and mask undercut along the features were the main problems. While local bowing was significantly reduced by trial and error,¹⁴ $\sim 5\ \mu\text{m}$ wide features still had a 0.65–1 μm mask undercut with walls angled at 88°. Requirements for mask undercut and sidewall angle in submicron features are more stringent. Blauw *et al.* reported similar etch rates for 30–50 μm wide features with aspect ratios as high as 25, but they also identified the crystal orientation dependence of etch rate as a factor that determines the feature profile at low temperatures.^{3,16}

^{a)}Author to whom correspondence should be addressed; electronic mail: aydil@engineering.ucsb.edu

In this article, we describe etching of deep submicron features at noncryogenic temperatures with a special emphasis on the relationship between etching figures of merit and several plasma parameters. The plasma parameters included F, O, and ion fluxes and the average energy of the ions impinging on the Si surface. The relative fluxes of ions and reactive neutrals in the plasma as well as the energy and angular distribution of ions impinging on the Si surface determine the rates of reactions that result in etching. However, these internal plasma parameters cannot be controlled directly during etching. Instead, externally controlled process variables such as pressure, gas flow rate, and composition, power supplied to generate the plasma, rf-bias power applied to the substrate electrode, and wafer temperature can be adjusted to control the internal plasma parameters, which in turn determine the process outcome. In this study, we aim to understand how the external process variables affect the process outcome through the key internal plasma parameters. Toward this end, we use plasma diagnostics in conjunction with etching experiments and examination of etched features through scanning electron microscopy (SEM).

II. EXPERIMENTAL SETUP AND PROCEDURE

Plasma etching experiments were conducted in a low-pressure, high-density inductively coupled plasma reactor (Lam Research TCP® 9400) similar to a reactor described previously.¹⁷ The plasma is maintained using a 13.56 MHz rf power source coupled through a matching network to a planar spiral coil seated on a quartz window. Wafers are placed on a chuck and electrostatically clamped with He backside cooling. The temperature of the anodized aluminum walls and the electrostatic chuck are maintained at 60 °C and 5 °C, respectively. Gas flow rates into the chamber are metered and controlled by mass flow controllers. Gases enter the reactor through an inlet near the quartz window and the chamber is pumped by a 1300 L/s turbomolecular pump. The typical operating pressure is between 5 and 80 mTorr and the pressure is controlled by a throttle valve. Wafers are loaded through a load-lock chamber to avoid frequent opening of the reactor to the atmosphere.

The electrostatic chuck (ESC) is powered by a second rf power supply, also at 13.56 MHz, to allow independent control of ion flux and ion energy. The current and voltage waveforms on the ESC are measured using a high-voltage probe and a coil that senses the rf current flowing in the connection to the electrode. The sheath potential drop determines the energy gained by the ions accelerated across the sheath toward the wafer surface. The sheath potential drop, $V_s(t)$, is the difference between the plasma potential $V_p(t)$, and the potential waveform at the electrode, $V_{rf}(t)$. Adjusting the rf-bias power applied to the electrostatic chuck changes $V_{rf}(t)$ and enables control of the ion energy. The rf-bias voltages reported in this article are the time-averaged voltage on the electrostatic chuck with respect to ground, $\langle V_{rf}(t) \rangle$. The time-averaged potential difference between the plasma and the electrostatic chuck represents the average energy of the ions impinging on the wafer surface and increas-

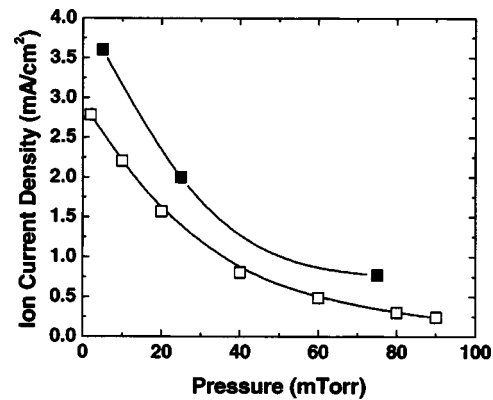


FIG. 1. Ion current density as a function of pressure in a SF_6 plasma as measured using an ion flux probe and as estimated from the rf-bias power and time-averaged rf-bias voltage. Lines are drawn through the data points to guide the eye.

ing the rf-bias voltage increases the average ion energy. The actual average ion energy is higher than $\langle V_{rf}(t) \rangle$ by $\langle V_p(t) \rangle$, which is on the order of 10–20 V (a few kT_e/e).

Various plasma diagnostics are available in the TCP reactor but only mass and optical emission spectroscopies and rf current and voltage probes were utilized in this study. The mass spectrometer was used to monitor stable species in the plasma and the effluent. Relative concentrations of reactive neutrals such as F and O were determined using optical emission spectroscopy (OES) in conjunction with actinometry.^{5,18–21} Optical emission from the plasma is collected and detected using an optical fiber coupled to a Verity SD 1024 Smart Detector Spectrograph with a 1024 element scientific grade sensor array. The detected emission is a spatial average across the discharge volume 10 mm above the surface of the electrostatic chuck. Optical emission spectra are collected between 200 nm and 800 nm and the resulting spectral resolution is better than 1.7 nm. Ar is added to the feed gas as the actinometer at 5% of the total flow rate. We use the F, O, and Ar emissions at 703.7, 777, and 750.4 nm, respectively, for actinometry. The positive ion flux impinging on the wafer surface is estimated from the rf-bias power, $\langle P_{rf} \rangle$ and the time-averaged rf-bias voltage, $\langle V_{rf} \rangle$ (i.e., self-dc bias) using $\Gamma_+ = \eta \langle P_{rf} \rangle / A \langle V_{rf} \rangle$, where A is the area of the electrostatic chuck and η is the fraction of the rf-bias power that goes into accelerating the ions across the sheath. We estimated η by comparing the above method to direct ion flux measurements conducted in a similar TCP reactor. Specifically, the positive ion flux impinging on the wafer surface was measured in pure SF_6 plasma using a small piece of heavily doped silicon mounted in the center of a 200 mm diameter heavily doped Si wafer with a Kapton tape placed in between itself and the wafer.²² The probe was negatively biased (–75 V) with respect to the wafer using a floating power supply and the probe current is determined from the potential drop across a precision resistor, measured using a battery-operated voltmeter. Figure 1 compares the ion flux estimated from $\langle P_{rf} \rangle$ and $\langle V_{rf} \rangle$ to the measurements made using the ion flux probe; η is $\sim 70\%$ at low pressures and

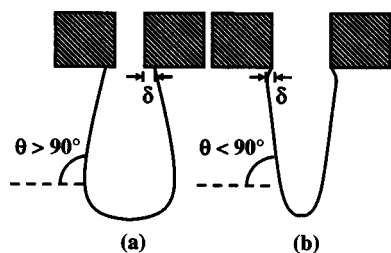


FIG. 2. Schematic showing mask undercut, δ , and sidewall slope, θ , parameters used to quantify the feature anisotropy. The feature shape is quantified through the mask undercut, δ , the lateral distance etched directly below the mask, and the sidewall slope, θ , the angle between the feature sidewall and the wafer plane. Sidewalls are said to be (a) negatively tapered when $\theta > 90^\circ$ and (b) positively tapered when $\theta < 90^\circ$.

decreases to 50% at higher pressures. While the absolute values of the ion flux may have systematic errors due to uncertainties in η , the trends in ion flux with varying plasma parameters are captured correctly.

The wafers used in the experiments are 200 mm diameter single crystal Si patterned with 0.35 and 0.5 μm diameter holes in a 1.2- μm -thick oxide mask deposited from tetraethoxysilane and O_2 plasma. The oxide mask was patterned through lithography using a photoresist mask. The pattern consists of repeated dies of same-size holes spaced 0.5 to 2.0 μm from each other. The holes in the oxide mask were etched using $\text{C}_4\text{F}_6/\text{O}_2$ plasma in a dual frequency capacitively coupled parallel plate reactor and the photoresist was subsequently removed using an O_2 discharge. Over 99% of the total wafer area is covered by the oxide mask and less than 1% of the Si wafer is exposed to the plasma.

All experiments were conducted using 800 W of rf power while varying the pressure, rf-bias voltage, total gas flow rate, and the SF_6 -to- O_2 ratio in the feed gas. Each wafer was etched for 150 s. Pressure and rf-bias voltage were varied from 5 to 75 mTorr and from zero to -160 V, respectively. Gas flow rate was varied from 40 to 250 sccm using various SF_6 -to- O_2 ratios. Etched wafers were cleaved and features from dies at the center of the wafer were examined using scanning electron microscopy (SEM). The time-averaged etch rate was calculated by dividing the etched depth with the etch time. Selectivity was determined from the ratio of the Si etch rate to the oxide mask etch rate. The mask undercut and sidewall slope were determined from the digitized SEM cross sections of the etched features and were used to quantify the feature anisotropy. The error on the undercut measurements is ± 5 nm and we estimate the error in the sidewall angle to be less than $\pm 0.5^\circ$. Specifically, we found it convenient to quantify the complex feature profile through two parameters: mask undercut, δ and sidewall angle (slope) θ . The mask undercut, δ , is the lateral etch observed directly below the oxide mask while the sidewall slope is quantified with the angle, θ , between the feature sidewall and the plane of the wafer surface, as shown in Fig. 2. Feature sidewalls are referred to as negatively tapered when they bow out toward the bottom of the feature with $\theta > 90^\circ$ [Fig. 1(a)], and as positively tapered when they slope inwards toward the bot-

tom of the feature with $\theta < 90^\circ$ [Fig. 1(b)]. Highly anisotropic features are those that have nearly no mask undercut and have approximately vertical sidewalls where $\theta = 90^\circ$.

III. RESULTS AND DISCUSSION

We systematically investigated the dependence of etch rate, selectivity, and the etched feature shape on the pressure, the rf-bias voltage and the SF_6 -to- O_2 ratio in the feed gas by adjusting these variables about their base values listed in Table I. The base values were arrived at after experiments designed to locate the center of the feasible and practical parameter space. The base conditions are also near the local optimum conditions that give a reasonable trade off between various requirements. In addition, we measured key plasma properties such as the ion flux, F concentration, and F-to-O flux ratio using plasma diagnostics. Below we discuss the effect of pressure, rf-bias voltage, and SF_6 -to- O_2 ratio on process figures of merit in light of the plasma diagnostic data. The total gas flow rate and total power were fixed to maximize the etch rate. The etch rate exhibits a maximum with flow rate at pressures > 10 mTorr; higher flow rates sweep out radicals faster than they can react to etch. The position of the maximum depends weakly on the gas composition and on the pressure.

A. Effect of pressure

Figures 3(a)–3(c) show the effect of pressure on the etch rate, selectivity, sidewall slope, and mask undercut. Figure 3(d) shows the ion current, F concentration, and F-to-O flux ratio in the plasma as a function of pressure. The Si etch rate exhibits a maximum value at 25 mTorr. At low pressures, where the ion flux is high, the etching is limited by the availability of F atoms. The etch rate increases with increasing pressure and F concentration up to the maximum. However, ion flux decreases with increasing pressure [Fig. 3(d)] and etching eventually becomes limited by the ion flux impinging on the surface. Average ion energy also decreases with increasing pressure as the sheath becomes increasingly collisional. The opposing trends in F and ion flux and energy with pressure give rise to an optimum pressure where the ion–neutral synergy is the greatest and the etch rate is maximum. The selectivity to the oxide mask also exhibits a maximum (> 50) at 25 mTorr. At low pressures, selectivity is low because the ion flux and, thus, the oxide-sputtering rate are high. At high pressures, the ion flux is low, but so is the Si etch rate, resulting in a low selectivity. The etch rate at 75 mTorr is very low and is nearly stopped because clearing of the oxide passivation by ion bombardment at the bottom of the feature is inadequate due to decreased ion energy and flux.

Figure 4 shows SEMs of 0.35 μm holes etched at 10, 25, 40, and 75 mTorr; the other conditions were kept constant at their base values listed in Table I. Mask undercut is worse at higher pressures. In fact, at 75 mTorr, the hole is hardly etched and we chose not to extract δ and θ values from the SEM. Since there is almost no ion bombardment on the sidewalls directly below the oxide mask, the observed lateral

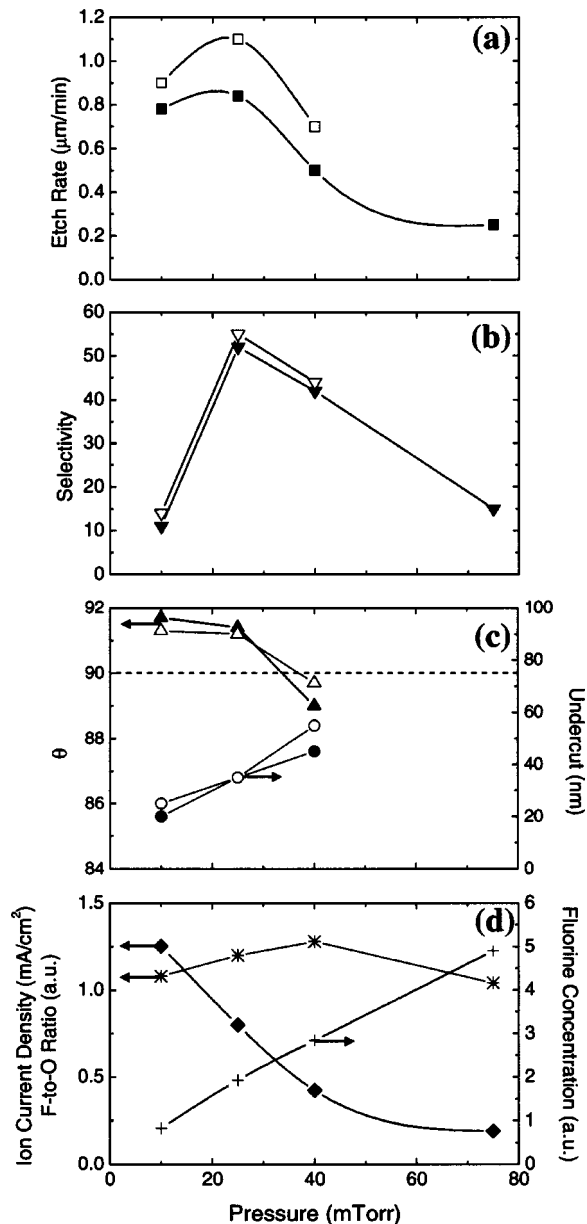


FIG. 3. (a) Si etch rate ($\blacksquare, \square$), (b) selectivity ($\blacktriangledown, \triangledown$), (c) mask undercut, δ ($\bullet, \circ$) and sidewall slope, θ ($\blacktriangle, \triangle$), and (d) ion current density ($\blacklozenge, \lozenge$), F-to-O ratio ($\ast$), and F concentration ($+$) as functions of pressure. Etch rate, selectivity, δ , and θ are shown for two different rf-bias voltages, -20 V (filled symbols) and -40 V (open symbols). Other plasma etching conditions were kept constant at the base values listed in Table I. Lines are drawn through the data points to guide the eye.

etching can only be due to chemical etching by F atoms coming to the surface as F or SF_x ($1 \leq x \leq 5$) fragments. As pressure is increased, the F concentration increases, resulting in high lateral etch rates, and more mask undercutting. The sidewall slope, however, is not sensitive to changes in pressure. As we will show shortly, the sidewall slope is determined by a F-to-O ratio, which remains approximately constant with increasing pressure. This is because both O and F concentrations increase approximately at the same rate with gas density. It is important to note that when the etched fea-

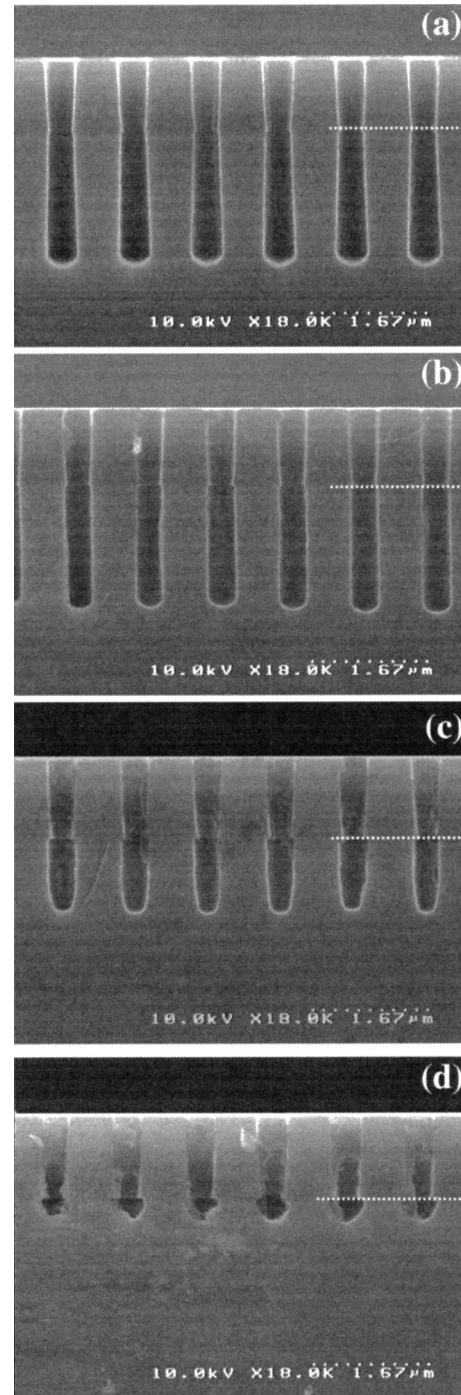


FIG. 4. SEM cross sections of holes etched at (a) 10 mTorr, (b) 25 mTorr, (c) 40 mTorr, and (d) 75 mTorr. Other plasma etching conditions were kept constant at the base values listed in Table I. The oxide mask-Si interface is shown with a dashed line.

tures are small, such as in Figs. 4(c) and 4(d), the sidewall slope is not well defined.

B. Effect of rf bias

Figures 5(a) and 5(b) show the effect of rf bias on the etch rate, selectivity, undercut, and the sidewall angle. In TCP plasma reactors, the rf-bias power applied to the electrostatic chuck controls the energy of the ions bombarding the wafer

TABLE I. Base etching process conditions.

Process variable	Base value	Range studied
Pressure	25 mTorr	5–75 mTorr
Total gas flow rate	80 sccm	
SF ₆ -to-O ₂ ratio in feed	1	0.6–2
TCP power	800 W	
rf-bias voltage	–20 V	0 to –160 V
Wafer temperature	5 °C	

surface independently from the flux of ions and reactive neutrals in the plasma.²³ Thus, ion and radical generation in the plasma is approximately decoupled from ion acceleration in the sheath and changing the rf-bias voltage primarily changes the ion energy. For example, Fig. 5(c) shows the ion current, F concentration, and F-to-O flux ratio as functions of the rf-bias voltage; indeed, these three plasma properties remain constant with changing rf-bias voltage. Increasing the ion energy increases the etch rate, but the increase is sublinear and eventually saturates at 1.7 $\mu\text{m}/\text{min}$ when the rf-bias increases above –120 V. Higher-energy ions also sputter the

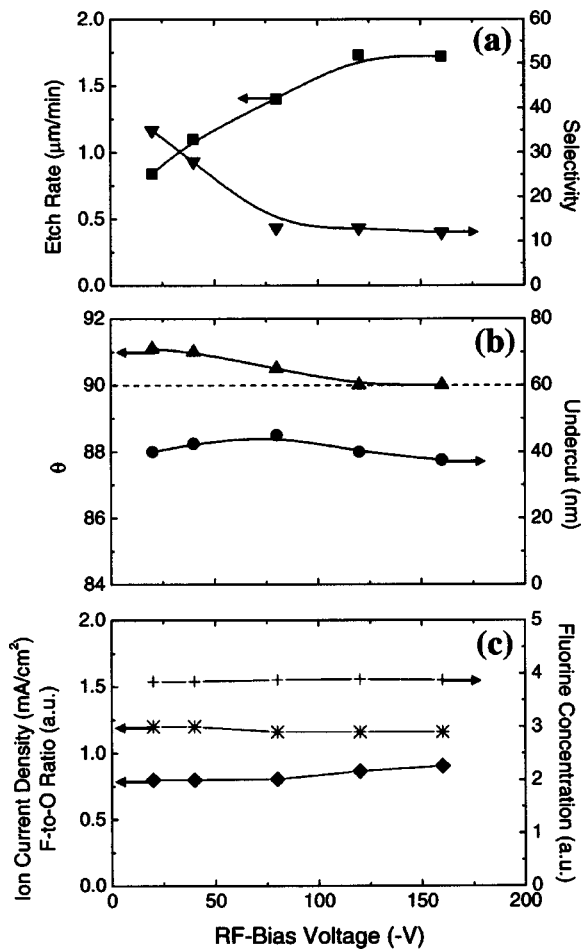


FIG. 5. (a) Si etch rate (■), (b) selectivity (▼), (c) mask undercut, δ (●), and sidewall slope, θ (▲), and (d) ion current density (◆), F-to-O ratio (*), and F concentration (+) as functions of rf-bias voltage. Other plasma etching conditions were kept constant at the base values listed in Table I. Lines are drawn through the data points to guide the eye.

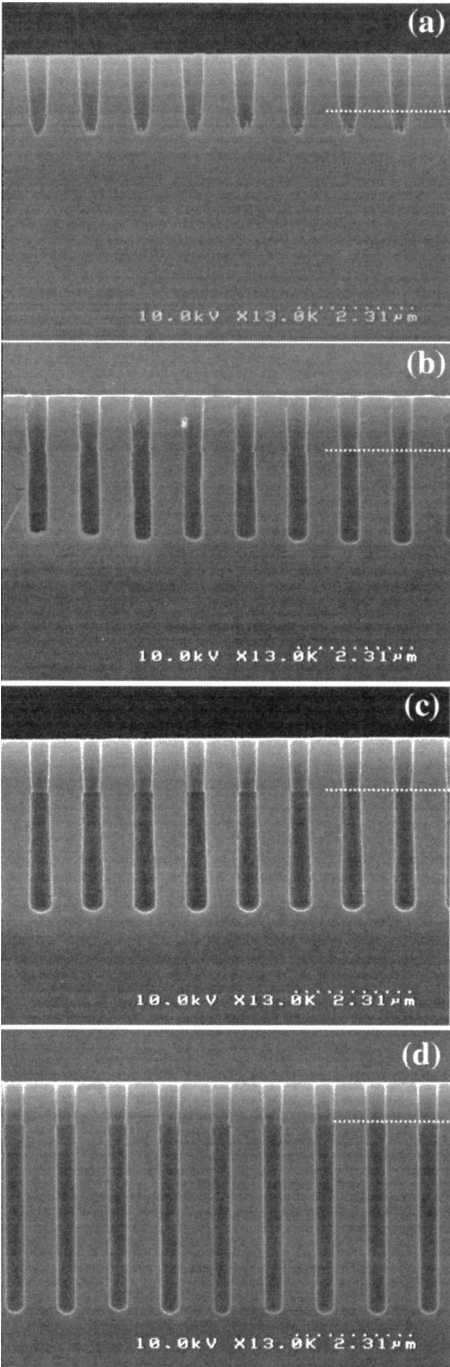


FIG. 6. SEM cross section of holes etched with (a) 0 V, (b) –20 V, (c) –40 V, and (d) –120 V rf-bias voltage applied to the electrostatic chuck. Other plasma etching conditions were kept constant at the base values listed in Table I. An oxide mask–Si interface is shown with a dashed line.

mask at a higher rate and selectivity decreases with increasing rf bias. Since the F concentration and F-to-O ratio remains constant, the lateral etch rate and undercut are independent of the rf-bias. The increase in the vertical etch rate relative to the lateral etch rate improves the anisotropy, as shown in Figs. 5(b) and 6.

C. Effect of SF₆-to-O₂ ratio

Figures 7(a) and 7(b) show the effect of SF₆-to-O₂ ratio, R , in the feed gas on the etch rate, selectivity, undercut and

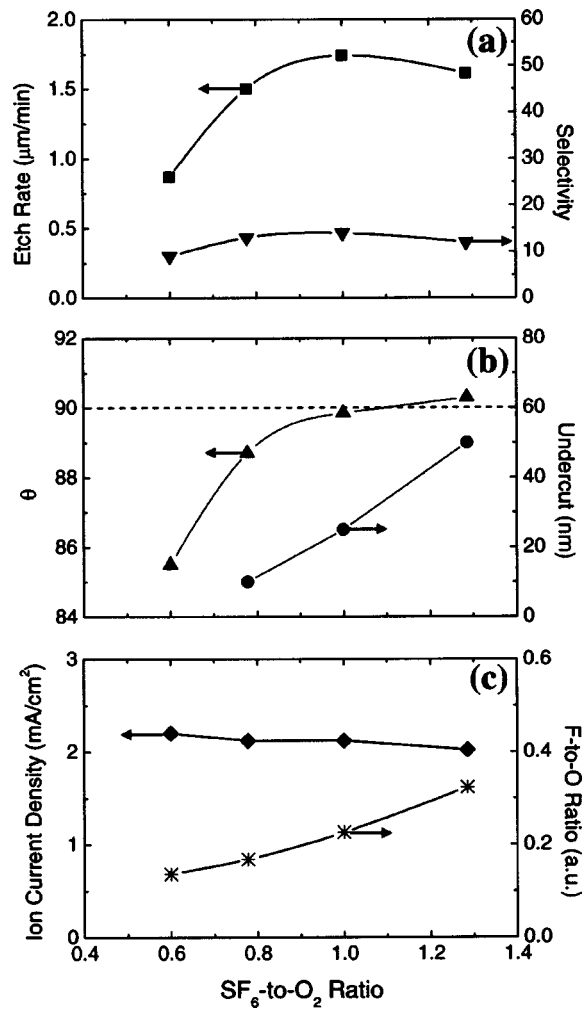


FIG. 7. (a) Si etch rate (■), (b) selectivity (▼), (c) mask undercut, δ (●), and sidewall slope, θ (▲), and (d) ion current density (◆), F-to-O ratio (*) as functions of the SF_6 -to- O_2 ratio in the feed gas. Other plasma etching conditions were kept constant at the base values listed in Table I. Lines are drawn through the data points to guide the eye.

the sidewall angle. Optical emission actinometry indicates that F-to-O ratio and F concentration in the plasma increase with R and, as a result, the etch rate increases. This etch rate increase is due to a higher F concentration in the plasma and is not due to higher ion flux, which remains remarkably constant over a wide range of R [Fig. 7(c)]. The mask-sputtering rate and, thus, the selectivity are also not affected by the SF_6 -to- O_2 ratio in the feed gas because the ion flux does not vary significantly with R .

Figure 7(b) shows that the mask undercut and the sidewall slope depend sensitively on the SF_6 -to- O_2 ratio in the feed gas. Small changes in R results in large changes in the feature shape (Figs. 8 and 9). At high SF_6 -to- O_2 ratios ($R > 1$), the mask undercut is large and the sidewalls are slightly negatively tapered [Figs. 7(b), 8(a), and 9]. The mask undercut is particularly severe when SF_6 -to- O_2 ratio is increased to 2 [Fig. 9(a)]. These effects can be understood in terms of the competitive adsorption of F and O atoms on the surface. The adsorption of O atoms leads to passivation, whereas adsorption of F leads to the etching of Si and any silicon oxide

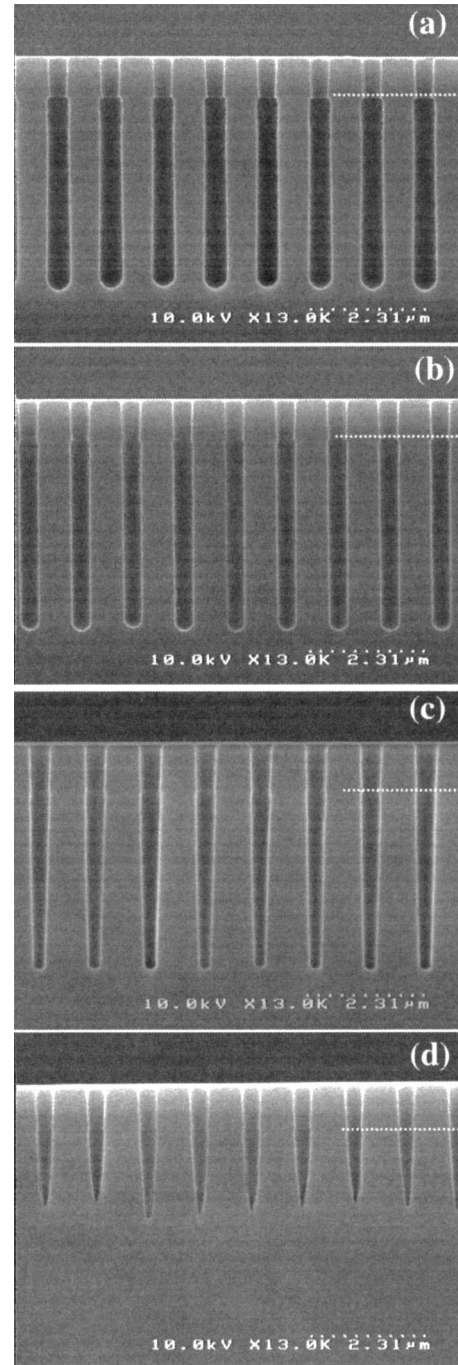


FIG. 8. SEM cross sections of holes etched with a SF_6 -to- O_2 ratio in the feed gas maintained at (a) 1.29 (45 sccm SF_6 /35 sccm O_2), (b) 1.00 (40 sccm SF_6 /40 sccm O_2), (c) 0.78 (35 sccm SF_6 /45 sccm O_2), and (d) 0.60 (30 sccm SF_6 /60 sccm O_2). The rf-bias voltage applied to the electrostatic chuck was -120 V and all other plasma etching conditions were kept constant at the base values listed in Table I. Oxide mask-Si interface is shown with a dashed line.

passivation layer that may form on the wall. At high R , sidewall passivation by O atoms adsorbing on the surface cannot keep up with the adsorption of F atoms, which can spontaneously etch Si, resulting in significant mask undercut. The negatively tapered sidewalls in Figs. 8(a) and 9(a) suggest that the ratio of F-to-O flux impinging on a particular posi-

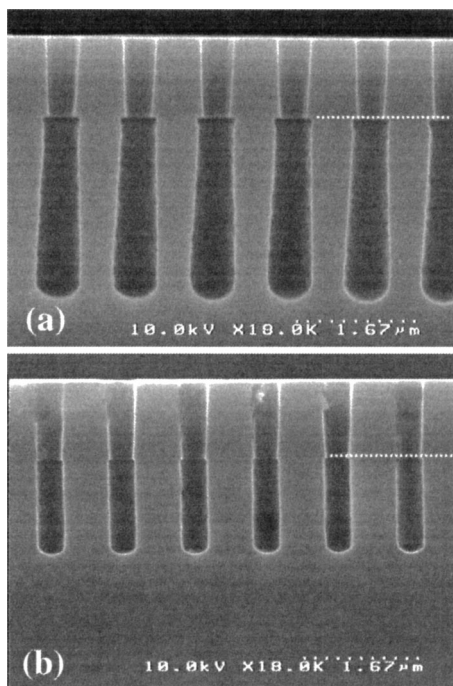


FIG. 9. SEM cross sections of holes etched with a SF_6 -to- O_2 ratio in the feed gas maintained at (a) 2 and (b) 1. The total feed gas flow rate was 250 sccm and all other plasma etching conditions were kept constant at the base values listed in Table I. The oxide mask-Si interface is shown with a dashed line.

tion in the feature increases with depth along the feature; sidewall passivation is less effective and lateral etching is faster at the bottom of the feature. This may be the case if the O sticking probability on Si is higher than that of F; fewer O atoms make it to the bottom of the feature because they adsorb, without reflection, on the sidewalls near the top of the feature. As the SF_6 -to- O_2 ratio in the feed gas is decreased to 1, effective sidewall passivation by O atoms results in less mask undercut and nearly vertical sidewalls [Fig. 8(b)]. In this case, competitive adsorption of O reduces F adsorption and Si etching. In addition, a very thin silicon oxide layer may form on the surface and F must etch this layer to reach the underlying Si adsorption sites. Little ion bombardment on the sidewalls combined with a low chemical etch rate of the oxide effectively reduces lateral etch rates. A further reduction in the SF_6 -to- O_2 ratio results in positively tapered sidewalls [Figs. 7(b), 8(c), and 8(d)].

D. Etching yield

We combined the etch rate, F , and ion flux measurements obtained at -20 and -40 V rf-bias voltage to produce a plot (Fig. 10) of the etching yield as a function of the F-to-ion flux (Γ_F/Γ_+) ratio. Figure 10 includes yields determined from all experiments conducted at pressures ranging between 10–40 mTorr and total flow rates between 40–250 sccm for an SF_6 -to- O_2 ratio of 1. Etching experiments at higher pressures were excluded because the average ion energy can be significantly lower than the rf-bias value due to collisions in the sheath.

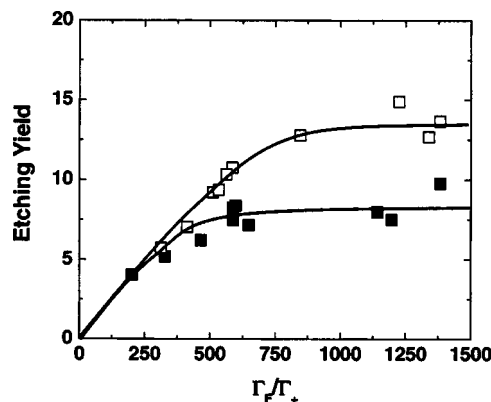


FIG. 10. Etching yield calculated from the experiments shown in Figs. 5 and 7 as a function of the F-flux-to-ion flux ratio, Γ_F/Γ_+ for two different rf bias (ion energy) values, -20 and -40 V. Lines are drawn through the data points to guide the eye.

The etching yield is defined as the number of Si atoms removed per incident ion impinging on the wafer surface, and is given by

$$Y = \frac{ER \cdot \rho_{\text{Si}}}{\Gamma_+}, \quad (1)$$

where ER is the etch rate and ρ_{Si} is the Si atomic density. Mass separation of the ions impinging on the chamber walls indicate that SOF_3^+ and SOF^+ are the dominant positive ions under the process conditions used in this study followed in magnitude by O_2^+ , SO^+ , SF_3^+ , and SF_5^+ , as shown in Fig. 11. The main etchant is assumed to be F atoms, but reactive SF_x fragments also impinge on the surface and are potential sources of F. By Γ_F , we mean the total flux of F atoms impinging on the surface as F as well as SF_x . We obtain an upper limit on Γ_F by estimating the degree of SF_6 dissociation in the plasma using mass spectrometry. The mass spectrum measured with no plasma (only gas) corresponds to the cracking pattern of SF_6 and O_2 in the mass spectrometer ionizer. When the plasma is turned on, ions created from SO_xF_y fragments are also detected, in addition to SF_x^+ , O^+ and O_2^+ . However, the SF_x^+ ions still have intensities consistent with the cracking pattern of SF_6 in the

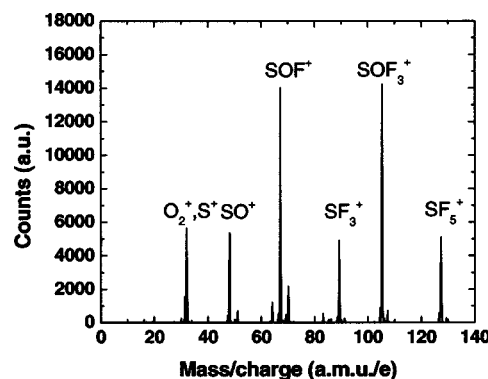


FIG. 11. Mass spectrum of positive ions in the plasma sampled through a pinhole in the chamber walls.

ionizer. This suggests that the majority of the SF_x species reaching the mass spectrometer is SF_6 . When the plasma is on, all SF_x^+ ion counts are much lower than those when the plasma is off, indicating that the degree of SF_6 dissociation is high ($\sim 80\%$). The SF_x fragments from the plasma recombine before reaching the spectrometer to form stable molecules such as SF_6 . We obtain an upper limit for Γ_F from the decrease in SF_6 by mass spectrometry, and assuming that all the F atoms from the dissociated SF_6 contribute to the etchant flux to the surface as F or SF_x . Thus, Γ_F values should be interpreted as the sum of F and SF_x ($1 \leq x \leq 5$) fluxes, $\Sigma_{i=F, SF_x} \Gamma_i$.

The yield curves as a function of the F-to-ion flux ratio have the same shape as the yield curves obtained in beam experiments where the Si surface is exposed to known fluxes of ion and radical (e.g., F) beams.²⁴ At a low Γ_F/Γ_+ ratio where etching is neutral-limited, the yield increases sharply with Γ_F/Γ_+ and eventually plateaus at high values of this ratio, where the etching becomes ion-limited. The transition between the neutral-limited and ion-limited regimes depends on the ion energy but occurs in the range where $\Gamma_F/\Gamma_+ \sim 10^3$. We are not aware of beam experiments, where the effects of fluorination with SF_x fragments or ion bombardment by SF_x^+ ions were investigated, but in the beam experiments conducted by Gray *et al.*, F atoms and Ar ions were used to study Si etching. At the same ion energy, the magnitudes of the etching yields shown in Fig. 10 are a factor of two higher than those reported by Gray *et al.*, and we attribute this difference to the differences in surface composition and microstructure when the Si surface is exposed to F atoms and Ar ions or a combination of F atoms, SF_x radicals and SF_x ions. In SF_6 plasma etching, the ions bombarding the surface are two to three times heavier than Ar^+ and are expected to result in somewhat thicker ion-mixing layers. Recently, Graves and Humbird showed through molecular dynamics simulations that some fluorine-containing ions such as F^+ and SiF_3^+ can create cracks and fissures in the ion mixing layer and increase the effective surface area for F adsorption and subsequent diffusion into the bulk.²⁵ Such a mechanism that enhances F mixing may also be operative in SF_6 etching, where ions such as SO_2F^+ , SF_3^+ , and SF_5^+ bombard the surface. In addition, the surfaces etched in the SF_6/O_2 plasma also receive O atom flux and O atoms compete with the surface sites. This, however, would tend to reduce the etching yield since etching yields for an oxidized silicon surface is significantly lower than that of Si. We should also note that using the uncorrected ion flux values from Fig. 1 brings the data in Fig. 10 in near perfect agreement with those of Gray. However, we believe that this agreement is fortuitous.

Despite the uncertainties in the compositions of the ion and radical fluxes, Fig. 10 indicates that the concepts from beam experiments can be extended and used to obtain yield data from a combination of plasma etching and plasma diagnostic experiments. Furthermore, analyzing the etching data in this manner helps process development and enables rational search for optimum process conditions. The effects of

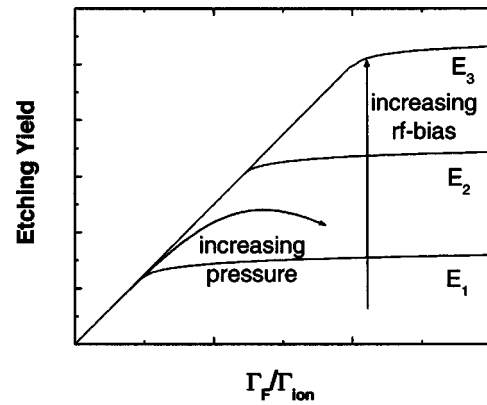


FIG. 12. A sketch of the trajectories, on the yield versus the Γ_F/Γ_+ plane, of the experiments where pressure and rf-bias voltage are changed. $E_1 < E_2 < E_3$ refer to ion energies.

process conditions can be understood in terms of the changes in the yield as a function of Γ_F/Γ_+ and ion energy. For example, the effect of changing the pressure and rf-bias on the etch rate can be understood by considering trajectories of these variations on the yield versus Γ_F/Γ_+ plane, as shown schematically in Fig. 12. Trajectories for a series of experiments where pressure or rf-bias voltage is increased while keeping all other parameters constant are shown on this plot. Increasing the reactor pressure results in an increase in the F flux and a decrease in the ion flux and ion energy. At higher pressure, the plasma potential is lower and the resulting sheath potential drop is also reduced. Thus, the ion energy decreases with increasing pressure. Furthermore, at higher pressures ions experience more collisions with neutrals while crossing the sheath and their energy decreases as a result of such collisions. Thus, as pressure is increased, the etching yield goes through a maximum and gives rise to the optimum etch rate in Fig. 3(a).

Increasing rf-bias (ion energy) while keeping the ion, F, and O fluxes constant is represented as a vertical line in the yield versus the Γ_F/Γ_+ plane. Increasing the rf-bias (ion energy) increases the etching yield, and, thus, the etch rate, but the rate of increase depends on the value of Γ_F/Γ_+ . In the region where etching is limited by ions, an increase in the ion energy results in a steady increase in the yield and thus the etching rate. Eventually, however, at some ion energy (rf bias) value, the vertical trajectory runs into the knee of the yield curve and etching becomes F-flux limited. At this point, the yield and the etch rate do not increase with a further increase in rf-bias. This is illustrated in Fig. 13, which shows the etch rate as a function of the rf-bias (ion energy) for various experiments where the Γ_F/Γ_+ was ~ 600 . Indeed, as rf bias (ion energy) is increased, etching eventually becomes F flux limited; for $\Gamma_F/\Gamma_+ \sim 600$ this happens at approximately -80 V.

In SF_6/O_2 plasma etching of Si, in addition to Γ_F , Γ_+ , and the ion energy, there is a fourth variable that affects the etching yield. This variable is the O atom flux, Γ_O . In generating and discussing Figs. 11 and 13, we isolated data for a single SF_6 -to- O_2 ratio (i.e., approximately constant F-to-O

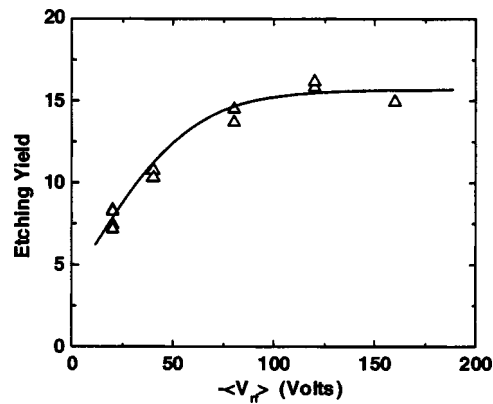


FIG. 13. Etching yield as a function of rf bias (ion energy) while keeping Γ_F/Γ_+ approximately constant at ~ 600 . (It is impossible to set the Γ_F/Γ_+ to a constant experimentally; the values used in generating this plot were distributed between 520 and 740. The mean was 600 and the standard deviation was 60.) The line through the data points was drawn to guide the eye.

ratio in the plasma) to eliminate the effects due to drastic changes in the nature of the surface; experiments with the same F-to-O ratio should yield surfaces with approximately the same ratio of F-to-O surface coverage. As the SF_6 -to- O_2 ratio in the feed gas is increased, the etching yield increases dramatically. For example, Table II compares etching yields for two values of the SF_6 -to- O_2 ratio in the feed gas. The etching yield approximately doubles when the SF_6 -to- O_2 ratio is increased by a factor of 2. We expect the F-to-O ratio to increase both on the silicon surface and in the ion-mixing layer when the SF_6 -to- O_2 ratio is increased. As expected from the relative etching yields of silicon dioxide and Si, the incorporation of O into the surface decreases the etching yield. The etching yields for an SF_6 -to- O_2 ratio of 2 with -20 V of rf bias are much larger than the etching yields reported by Gray *et al.* for 20 eV Ar ions (<5) on Si surfaces exposed to F atoms. Evidently, fluorination by SF_6 fragments and SF_x^+ ions results in much higher yields.

IV. SUMMARY AND CONCLUSIONS

In SF_6/O_2 plasma etching of deep holes in Si, externally controllable process variables such as the pressure, rf-bias voltage and the SF_6 -to- O_2 ratio in the feed gas determine the Si etch rate, selectivity, feature shape, and anisotropy through their effects on F, O, and ion fluxes as well as the

TABLE II. A comparison of etching yields for two different SF_6 -to- O_2 ratios in the feed gas.

SF_6 -to- O_2 ratio	$\langle V_{rf} \rangle$ (V)	Γ_F/Γ_+	Etching yield
1	-20	650	7 ± 0.5
1	-20	1200	7.5 ± 0.5
1	-20	1100	8 ± 0.5
1	-20	2400	7 ± 0.5
2	-20	800	9 ± 0.5
2	-20	1300	13 ± 0.5
2	-20	2600	14 ± 0.5

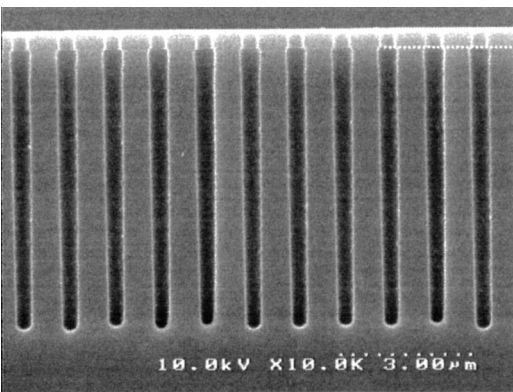


FIG. 14. The SEM cross section of $0.35\text{-}\mu\text{m}$ holes etched under plasma operating conditions listed in Table III. The oxide mask-Si interface is shown with a dashed line.

energy of the ions impinging on the surface. The pressure determines the relative fluxes of F and ions impinging on the surface, which in turn determines the etch rate and selectivity. Increasing the pressure increases the F flux but decreases the ion flux. Thus, etching is neutral-limited at low pressures but becomes ion-limited at high pressures. The synergy between ions and neutrals results in a maximum etch rate at intermediate pressures (~ 25 mTorr). The selectivity also exhibits a maximum value with pressure: high ion flux at low pressure results in high oxide mask sputtering rates and low ion flux at high pressures translates to low Si etch rates. Mask undercutting, however, worsens with increasing pressure because the higher F-to-O flux ratio results in higher lateral etch rates and less anisotropic features. Increasing the rf-bias voltage increases the average ion energy and increases the vertical etch rate relative to the lateral etch rate and the feature anisotropy improves. However, the selectivity worsens with increasing rf-bias due to an increased mask sputtering rate. The SF_6 -to- O_2 ratio in the feed gas determines the F-to-O flux ratio in the plasma, which controls the etch rate as well as the feature shape and anisotropy. The selectivity, however, is not affected by this ratio. Increasing the SF_6 -to- O_2 ratio increases the F-to-O flux ratio in the plasma and, thus, increases the etch rate, but compromises sidewall passivation. At a high SF_6 -to- O_2 ratio, the mask undercut is severe and the sidewalls are negatively tapered. As the SF_6 -to- O_2 ratio is decreased, effective sidewall passivation by O atoms results in less mask undercut and nearly vertical sidewalls. However, a further reduction in SF_6 -to- O_2 ratio results in low etch rates and positively tapered sidewalls.

Figure 14 is a cross section of $0.35\text{ }\mu\text{m}$ holes etched using the process conditions listed in Table III that gives an acceptable combination of high etch rate, selectivity, and feature anisotropy. The total etch depth is $7\text{ }\mu\text{m}$, which corresponds to an aspect ratio of 20. The time-averaged etch rate is $1.3\text{ }\mu\text{m/min}$ and the selectivity to the oxide mask is 9. The feature is highly anisotropic with nearly vertical sidewalls ($\theta \sim 90^\circ$). However, there is still some observable mask undercut, which may not be acceptable for features smaller than

TABLE III. Process conditions that result in the feature profile shown in Fig. 13.

Process Variable	Value
Pressure	25 mTorr
Total gas flow rate	80 sccm
SF ₆ -to-O ₂ ratio in feed	1
TCP power	800 W
rf-bias voltage	−120 V
Wafer temperature	5 °C

0.1 μm . Herein we identified the effects of important process parameters and trade-offs in SF₆/O₂ plasma etching of high aspect ratio features. Future optimizations for smaller holes and trenches maybe based on the framework established herein.

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